

Exact Las Vegas query costs for purification with state-generating oracles

Abstract

Let a unitary oracle prepare a state from a known input, with probability p of measuring answer 1, where $|p - 1/2| \geq \delta$ for a known $0 < \delta < 1/2$. We determine the minimax Las Vegas query cost of returning the more likely answer exactly in the transducer model, where each query is charged the squared norm of its queried component. Arbitrary unitary completions and potentially infinite-dimensional separable private space are allowed. With unrestricted controlled access to the oracle and its inverse, the optimum is $1/(2\delta)$, removing the additive unit in the bound of Belovs and Jeffery (Quantum 2026). If forward queries are confined to the known input ray and inverse queries to the generated-state ray, the optimum is $1/(1 - \sqrt{1 - 4\delta^2})$, asymptotic to $1/(2\delta^2)$ as $\delta \rightarrow 0$. For every nonzero finite-dimensional oracle workspace, each optimum is attained by a transducer fixed over the whole promised family. The unrestricted construction pulls the answer reflection back to the known input without requiring knowledge of the resulting invariant plane; a four-oracle adversary proves optimality. For restricted access, a signed-Gram characterization proves the optimality of the earlier reciprocal-rescaling construction.

Note. This paper was generated entirely by AI, including MiMo, using an automated research pipeline developed by Chenghua Liu and Hanyu Li.

1 Introduction

A bounded-error quantum subroutine supplies more than a sample of its answer. Its unitary implementation can be reversed and, in principle, called on any input. Purification asks how to use this coherent access to return the subroutine's more likely answer exactly. We study Boolean purification in the transducer model, where an exact public transformation is accompanied by an unchanged auxiliary vector in a private direct-sum space. The associated Las Vegas cost charges the squared norm processed by each oracle call. This is the resource that accounts for the use of a subroutine on a particular state under composition [2, 3].

The distinction between a state and a unitary is important even for this one-bit task. Write $Os = \phi_O$, where s is a known input and measuring the answer qubit of ϕ_O returns 1 with probability p_O . We are promised that $|p_O - 1/2| \geq \delta > 0$. The workspace may retain arbitrary garbage, and the action of O on $s^\perp$ is unspecified. A purifier must therefore work for every unitary having the prescribed state-generation property. Queries away from s can be useful, but an algorithm cannot choose a convenient completion and treat its action as known.

Belovs and Jeffery constructed an optimal reflecting-oracle purifier of cost $1/(2\delta)$ and obtained cost $1 + 1/(2\delta)$ for state-generating access [4, Theorem 1.5 and Corollary 1.6]. The additive unit has a concrete origin. Their reduction prepares ϕ_O on one branch of an equal superposition, applies the phase purifier, and uncomputes ϕ_O . The preparation and uncomputation each process squared norm $1/2$. We remove these two calls by basing the purifier at the known input s . The optimality of the reflecting-oracle primitive by itself does not show that this extra unit is necessary for the state-generating task.

There is also a more restrictive way to expose the subroutine: permit forward queries only on the input ray $\mathbb{C}s$, and inverse queries only on the generated ray $\mathbb{C}\phi_O$. This interface permits state

generation and uncomputation, with coherent ancillary controls, but not queries on other directions. Belovs and Jeffery's Open Problem B asks for the exact state-generating value and a restricted-query variant. We resolve the unrestricted question and determine the exact value for this bidirectional ray restriction. In particular, the older quadratic-in- $1/\delta$ restricted constructions cannot be improved to linear cost within this interface.

For a nonzero finite-dimensional workspace W , let $C_{\text{sg},W}(\delta)$ and $C_{\text{adm},W}(\delta)$ denote the minimax Las Vegas costs with unrestricted and ray-restricted access, respectively. Both allow controlled O and O^* , including a non-query branch. The minimization is over one fixed transducer that serves all promised unitaries, and the maximum is over those unitaries and their completions. Section 2 gives the formal definitions.

Theorem 1.1. *For every nonzero finite-dimensional W and every $0 < \delta < 1/2$,*

$$C_{\text{sg},W}(\delta) = \frac{1}{2\delta}, \quad (1.1)$$

$$C_{\text{adm},W}(\delta) = \frac{1}{1 - \sqrt{1 - 4\delta^2}} = \frac{1 + \sqrt{1 - 4\delta^2}}{4\delta^2}. \quad (1.2)$$

Both infima are attained using separable private space.

The first equality removes the preparation overhead without changing the oracle specification. The second identifies the exact cost of restricting query support. For example, when $\delta = 1/4$, the two costs are 2 and $4 + 2\sqrt{3}$, while the previous unrestricted upper bound is 3. As the gap vanishes,

$$C_{\text{adm},W}(\delta) = \frac{1}{2\delta^2} - \frac{1}{2} + O(\delta^2), \quad C_{\text{sg},W}(\delta) = \frac{1}{2\delta}.$$

Neither value depends on $\dim W$. These exact optima concern transduction with potentially infinite-dimensional private space, rather than finite-query zero-error algorithms in the ordinary circuit model.

To avoid preparing ϕ_O , we pull the known answer reflection back through O :

$$R_O = O^*(2\Pi_0 - I)O,$$

where Π_0 projects onto answer 0. Its expectation at the fixed input is $\langle s, R_O s \rangle = 1 - 2p_O$, so the desired answer is encoded by the sign of this quantity. The span of s and $R_O s$ is invariant and has dimension at most two. Merely identifying that plane is not enough: its second basis vector depends on the unknown completion, and a change of basis using that vector would not be a legal work operation. We implement the line purifier of [4, Section 3.1] with counter shifts controlled only by $|s\rangle\langle s|$ and its complement. Thus its restriction to each unknown plane has the required two-dimensional action, while the circuit itself is fixed. The two oracle calls implementing each R_O are offset by the factor $1/2$ from converting phase to bit; there is no separate preparation or uncomputation call.

The corresponding lower bound also has to see the entire oracle. Two boundary rotations distinguish the answer classes, but their adversary filter gives a constant strictly below $1/(2\delta)$. We include the negative of each active rotation, which preserves its answer probability. A joint adversary on these four unitaries has filter norm exactly 2δ . This proves the unrestricted optimum already on one answer qubit, and embeds into every workspace.

Under the ray restriction the completion instead drops out of the adversary identity. Its two query directions contribute opposite signs, leaving a difference of Gram matrices multiplied by $1 - \langle \phi_x, \phi_y \rangle$. The earlier reciprocal rescaling of the answer blocks [1, Section 6.2] [3, Section 13.1] preserves

cross-class inner products while reducing every squared norm to at most $\sqrt{1 - 4\delta^2}$. Summing tensor powers then realizes the reciprocal kernel with exactly the cost in (1.2). A boundary pair of states attains the same overlap and gives a matching lower bound. This is an exact analysis of the existing rescaling mechanism. We also show why the two directions cannot be separated: using only one leaves same-label witnesses mutually orthogonal, forcing unbounded uniform cost (Proposition 4.2).

The proofs use one adversary identity, developed in Section 2. Its converse includes a unitary-extension argument for the whole oracle family, so the restricted witnesses define one transducer even though the family is uncountable. Sections 3 and 4 prove the two equalities.

2 Query model and adversary realizations

2.1 Oracle access and exact transduction

Let $A \cong \mathbb{C}^2$, let $W \neq \{0\}$ be a finite-dimensional Hilbert space with a designated unit vector $|0_W\rangle$, and put

$$M = A \otimes W, \quad s = |0_A 0_W\rangle, \quad \Pi_b = |b\rangle\langle b|_A \otimes I_W \quad (b \in \{0, 1\}).$$

An input is a unitary $O: M \rightarrow M$. Write

$$\phi_O = Os, \quad p_O = \|\Pi_1 \phi_O\|^2. \quad (2.1)$$

For a fixed $0 < \delta < 1/2$, the promised family $\mathcal{O}_{\delta, W}$ consists of all such unitaries with

$$p_O \leq \frac{1}{2} - \delta \quad \text{or} \quad p_O \geq \frac{1}{2} + \delta. \quad (2.2)$$

Let $r_O = 0$ in the first case and $r_O = 1$ in the second. When both answer blocks are nonzero, the generated state can be written as

$$\phi_O = \sqrt{1 - p_O} |0\rangle \phi_{O,0} + \sqrt{p_O} |1\rangle \phi_{O,1},$$

where the two garbage states are normalized. They may be complex, nonorthogonal, and unrelated across instances. A garbage state multiplying a zero coefficient can be chosen arbitrarily. The action of O on $s^\perp$ is unrestricted.

We use the controlled arbitrary-unitary query convention of [1, 2, 4]. The bidirectional oracle is

$$Q_O = O \oplus O^*, \quad (2.3)$$

with orthogonal forward and inverse direction labels. A query applies

$$\tilde{Q}_O = I_{\text{nonquery}} \oplus (I \otimes Q_O).$$

Thus the circuit may coherently choose whether to query and in which direction. Oracles are specified as unitaries, not modulo global phase; a phase of an oracle block is observable relative to a non-query branch. Work operations are arbitrary fixed unitaries and are not charged to the query cost.

The public space is $\mathcal{H} = \mathbb{C}_B^2 \otimes \mathbb{C}s$, with basis $\xi_b = |b_B\rangle s$. An exact purifier is a fixed query circuit S on $\mathcal{H} \oplus \mathcal{L}$, together with a choice of catalyst $w_O \in \mathcal{L}$ for every promised oracle, satisfying

$$S(Q_O)(\xi_0 \oplus w_O) = \xi_{r_O} \oplus w_O. \quad (2.4)$$

The circuit has finitely many query gates. Its private space $\mathcal{L}$, query controls, and work unitaries are independent of the unknown oracle; w_O may depend on it. Catalysts here are unnormalized

vectors in a *direct sum*, not auxiliary tensor-product states supplied to an ordinary algorithm. A catalyst is included in the specification of the realization, as it need not be unique [3, Section 7.1].

For such a realization $\mathcal{T} = (\mathbf{S}, (w_O)_O)$, let $q_{O,j}$ be the queried component immediately before its j -th query, when the circuit acts on $\xi_0 \oplus w_O$. Its Las Vegas query cost and its transduction complexity are, respectively,

$$L_{\mathcal{T}}(O) = \sum_j \|q_{O,j}\|^2, \quad \|w_O\|^2. \quad (2.5)$$

Private space may be infinite-dimensional. The constructions below use separable space, with arbitrary fixed work unitaries. Work operations are not subject to a finite-gate synthesis requirement.

For the restricted interface, stack the query vectors using orthogonal query-time labels. We call the realization *admissible* if, for every O , its total query vector has the form

$$v_O = (f_O \otimes s)_{\text{fwd}} \oplus (g_O \otimes \phi_O)_{\text{inv}}. \quad (2.6)$$

Thus every forward query is supported on the designated input ray and every inverse query on the generated-state ray. The vectors f_O, g_O contain all remaining controls and registers. Different control spaces can be embedded in one common control space. After the query,

$$Q_O v_O = (f_O \otimes \phi_O)_{\text{fwd}} \oplus (g_O \otimes s)_{\text{inv}}, \quad (2.7)$$

where identity operators on controls are suppressed. Define

$$C_{\text{sg},W}(\delta) = \inf_{\mathcal{T} \text{ exact}} \sup_{O \in \mathcal{O}_{\delta,W}} L_{\mathcal{T}}(O), \quad (2.8)$$

$$C_{\text{adm},W}(\delta) = \inf_{\mathcal{T} \text{ exact and admissible}} \sup_{O \in \mathcal{O}_{\delta,W}} L_{\mathcal{T}}(O). \quad (2.9)$$

In both infima, one fixed circuit must serve the entire promised family.

2.2 The adversary identity and its converse

Throughout, inner products are conjugate-linear in the first argument. Consider any exact state-conversion family $(\xi_x, \tau_x, Q_x)_{x \in X}$, with one fixed transducer and catalysts w_x . If $q_{x,j}$ denotes the component queried at step j , telescoping inner products through the circuit gives

$$\langle \xi_x, \xi_y \rangle - \langle \tau_x, \tau_y \rangle = \langle v_x, (I - Q_x^* Q_y) v_y \rangle, \quad v_x = \bigoplus_j q_{x,j}. \quad (2.10)$$

Here and below, oracle operators act identically on ancillary and query-time controls. To see the identity, an input-independent work unitary preserves each cross-instance inner product. A query changes that inner product from its pre-query value by $-\langle q_{x,j}, (I - Q_x^* Q_y) q_{y,j} \rangle$. Summing these changes and subtracting the initial and final inner products cancels $\langle w_x, w_y \rangle$, because the catalyst is returned. Orthogonality of the query-time labels then gives (2.10), and $\|v_x\|^2$ is the query cost. This is the transducer form of the adversary identity [2, 4].

For the restricted upper bound we will construct query vectors directly from the generated states. To turn those vectors into a single circuit, we need the converse of (2.10) for the entire oracle family. Equal Gram matrices define an isometry on the closed spans, but in infinite dimension their orthogonal complements need not have equal dimension. An unused summand removes this obstruction.

Lemma 2.1 (Stabilized Gram realization). *Let $\mathcal{V}$ be a Hilbert space and let $(a_x)_{x \in X}, (b_x)_{x \in X} \subseteq \mathcal{V}$ satisfy $\langle a_x, a_y \rangle = \langle b_x, b_y \rangle$ for all x, y . Set $\varkappa = \max\{\aleph_0, \dim \mathcal{V}\}$, where dimension means Hilbert-space dimension. There is a unitary U on $\mathcal{V} \oplus \ell^2(\varkappa)$ such that*

$$U(a_x \oplus 0) = b_x \oplus 0 \quad (x \in X).$$

In particular, a separable $\mathcal{V}$ requires only a separable added summand, regardless of the cardinality of X .

Proof. The assignment $\sum_x \alpha_x a_x \mapsto \sum_x \alpha_x b_x$, for finitely supported coefficient families, is well defined and preserves norms by the Gram identities. It extends to a surjective isometry between the two closed spans. After adjoining $\ell^2(\varkappa)$, the orthogonal complement of either span has dimension $\varkappa$: it contains the new summand and its dimension is at most $\varkappa$. Choose a unitary between these complements and take its orthogonal direct sum with the isometry of the closed spans. $\square$

If query vectors v_x satisfy (2.10), the two families

$$a_x = \xi_x \oplus Q_x v_x, \quad b_x = \tau_x \oplus v_x$$

have equal Gram matrices. The lemma supplies a fixed unitary U with

$$U(\xi_x \oplus Q_x v_x \oplus 0) = \tau_x \oplus v_x \oplus 0. \quad (2.11)$$

Applying the oracle on the query summand and then U gives a canonical transducer. Its catalyst is $v_x \oplus 0$, so both its query cost and its transduction complexity are $\|v_x\|^2$. The added summand is non-query space and is unpopulated on every instance. This proves the needed converse without assuming finiteness of X ; compare [4, Theorem 2.15].

Remark 2.2. An arbitrary infinite-dimensional unitary need not admit a transduction on every public input. For example, let $V e_n = e_{n+1}$ on $\ell^2(\mathbb{Z})$, and declare $\mathbb{C} e_0$ public. The equation $V(e_0 + w) = \tau e_0 + w$, with $w \perp e_0$, forces the coefficient of e_n in w to be one for every $n \geq 1$, contrary to square summability. Our upper bounds instead either provide the catalyst explicitly or construct the work unitary from the entire equal-Gram family using Lemma 2.1.

3 Unrestricted queries

3.1 A purifier based at the known input

We first construct a phase purifier whose public input is s . The construction has two parts: a reflection whose compression to the plane generated by s records the answer probability, and a line walk that uses this plane without requiring its basis as input. Define

$$Z_A = 2\Pi_0 - I_M, \quad P_O = O^* \Pi_0 O, \quad R_O = 2P_O - I_M = O^* Z_A O. \quad (3.1)$$

Each controlled call to R_O is implemented by a controlled O query, the known reflection Z_A on the same branch, and a controlled O^* query. The two oracle queries process equal squared norms.

Fix an oracle and abbreviate $p = p_O$. For $0 < p < 1$, put

$$a = \sqrt{1-p}, \quad b = \sqrt{p}, \quad u_0 = \frac{P_O s}{a}, \quad u_1 = \frac{(I - P_O)s}{b}.$$

The identity $\|P_O s\|^2 = \langle Os, \Pi_0 Os \rangle = 1 - p$, and its counterpart for $I - P_O$, show that u_0, u_1 are orthonormal. They are respectively $+1$ and -1 eigenvectors of R_O , and $s = au_0 + bu_1$. Consequently,

$$t_O = bu_0 - au_1 \quad (3.2)$$

is a unit vector orthogonal to s . The plane $E_O = \text{span}\{s, t_O\}$ is invariant, and in its ordered basis (s, t_O) ,

$$R_O|_{E_O} = \begin{pmatrix} 1 - 2p & 2\sqrt{p(1-p)} \\ 2\sqrt{p(1-p)} & 2p - 1 \end{pmatrix} = 2|\psi_p\rangle\langle\psi_p| - I_{E_O}, \quad \psi_p = as + bt_O. \quad (3.3)$$

Equation (3.3) isolates the completion dependence: the matrix is determined by p , but its basis vector t_O is not. The rank of P_O and the garbage states do not enter that matrix. The next lemma implements the required walk using the projector onto s , so no gate is defined by t_O .

Lemma 3.1 (Line purifier on an unknown invariant plane). *Let s be a fixed unit vector. Suppose a reflection R has an invariant plane $E = \text{span}\{s, t\}$, where s, t are orthonormal and*

$$R|_E = 2|\sqrt{1-p}s + \sqrt{p}t\rangle\langle\sqrt{1-p}s + \sqrt{p}t| - I_E, \quad 0 < p < 1, \quad p \neq \frac{1}{2}.$$

There is a fixed transducer, independent of p and t , whose operations depend only on s and controlled access to R , and whose public action is

$$s \mapsto \begin{cases} s, & p < 1/2, \\ -s, & p > 1/2. \end{cases} \quad (3.4)$$

Its reflection-query cost and the squared norm of its catalyst are

$$\frac{1}{|1 - 2p|}, \quad \frac{\min\{p, 1 - p\}}{|1 - 2p|}, \quad (3.5)$$

respectively. The same circuit has zero catalyst and cost one when $Rs = s$ or $Rs = -s$, with the corresponding public sign.

Proof. We realize the weighted line purifier of [4, Section 3.1] using shifts controlled only by the known projector $P_s = |s\rangle\langle s|$. Let $J = \ell^2(\mathbb{Z})$, let $S_J|j\rangle = |j+1\rangle$, and define

$$T_s = S_J \otimes P_s + I_J \otimes (I - P_s), \quad T_\perp = I_J \otimes P_s + S_J \otimes (I - P_s).$$

Both are unitaries. With $P_0 = |0\rangle\langle 0|_J$, set

$$\mathcal{R}_1 = T_s^* [P_0 \otimes I + (I_J - P_0) \otimes R] T_s, \quad (3.6)$$

$$\mathcal{R}_2 = T_\perp^* [P_0 \otimes I + (I_J - P_0) \otimes (-R)] T_\perp, \quad \mathcal{P}_R = \mathcal{R}_2 \mathcal{R}_1. \quad (3.7)$$

The shifts move the s and $s^\perp$ components separately; on any invariant plane they therefore pair consecutive sites of a line. The conditional minus sign is oracle-independent. The public line is $\mathbb{C}(|0\rangle s)$; its orthogonal complement in the fixed space $J \otimes M$ is private.

Put $\gamma = \sqrt{p/(1-p)}$, and define orthonormal vectors

$$\begin{aligned} e_{4k} &= |4k\rangle s, & e_{4k+1} &= |4k+1\rangle t, \\ e_{4k+2} &= -|4k+2\rangle s, & e_{4k+3} &= -|4k+3\rangle t \end{aligned} \quad (k \geq 0).$$

The closed span $\mathcal{K}_+ = \overline{\text{span}}\{e_j : j \geq 0\}$ is invariant under both reflections. More explicitly, conjugating by the shifts in (3.6)–(3.7) gives, in the indicated ordered bases,

$$\mathcal{R}_1|_{\text{span}\{e_{2k}, e_{2k+1}\}} = \mathcal{R}_2|_{\text{span}\{e_{2k+1}, e_{2k+2}\}} = B_\gamma, \quad B_\gamma = \frac{1}{1+\gamma^2} \begin{pmatrix} 1-\gamma^2 & 2\gamma \\ 2\gamma & \gamma^2-1 \end{pmatrix}, \quad (3.8)$$

and $\mathcal{R}_2 e_0 = e_0$. These blocks exhaust $\mathcal{K}_+$, with e_0 the only unpaired vector for $\mathcal{R}_2$. The matrix B_γ fixes $(1, \gamma)$ and negates $(1, -\gamma^{-1})$.

If $\gamma < 1$, choose

$$v = \sum_{j \geq 1} \gamma^j e_j.$$

Every paired coefficient vector of $e_0 + v$ is proportional to $(1, \gamma)$. Both reflections fix the coupling, so $\mathcal{P}_R(e_0 + v) = e_0 + v$. If $\gamma > 1$, choose instead

$$v = \sum_{j \geq 1} (-\gamma)^{-j} e_j.$$

Now every pair is proportional to $(1, -\gamma^{-1})$. The first reflection sends $e_0 + v$ to $-e_0 - v$. The second fixes the boundary component $-e_0$ and negates all remaining pairs, giving $\mathcal{P}_R(e_0 + v) = -e_0 + v$. The series converge in norm, with

$$\|v\|^2 = \begin{cases} \frac{\gamma^2}{1-\gamma^2} = \frac{p}{1-2p}, & p < 1/2, \\ \frac{\gamma^{-2}}{1-\gamma^{-2}} = \frac{1-p}{2p-1}, & p > 1/2. \end{cases} \quad (3.9)$$

At the first query, T_s has moved every populated component to a nonzero counter value, so the queried squared norm is $1 + \|v\|^2$. Before the second query, $T_\perp$ leaves only the boundary component at counter zero; the queried squared norm is $\|v\|^2$. Thus the total cost is $1 + 2\|v\|^2 = 1/|1-2p|$. The basis (e_j) and the catalyst depend on t , but the circuit and the public/private decomposition do not. Finally, if $Rs = \pm s$, $v = 0$ gives $\mathcal{R}_1 e_0 = \pm e_0$ and $\mathcal{R}_2 e_0 = e_0$. Only the first controlled query is populated, with squared norm one. $\square$

Apply the lemma to R_O and (3.3). The cases $p_O = 0, 1$ satisfy $R_O s = s, -s$, respectively, and are covered by the last part of the lemma. The resulting phase transducer therefore works for every promised completion. Its circuit uses only P_s , counter shifts, and the pulled-back reflection; the gap is needed only to bound the worst-case cost.

To turn the phase into a bit, take the direct sum of the identity on $\mathbb{C}\xi_0$ and the phase transducer on $\mathbb{C}\xi_1 \oplus \mathcal{L}$. Conjugate this operation by a Hadamard on the *public* space $\text{span}\{\xi_0, \xi_1\}$, extended as the identity on $\mathcal{L}$. For the phase catalyst v_O , the resulting transformation is

$$\begin{aligned} \xi_0 \oplus \frac{v_O}{\sqrt{2}} &\mapsto \frac{\xi_0 + \xi_1}{\sqrt{2}} \oplus \frac{v_O}{\sqrt{2}} \\ &\mapsto \frac{\xi_0 + (-1)^{r_O} \xi_1}{\sqrt{2}} \oplus \frac{v_O}{\sqrt{2}} \mapsto \xi_{r_O} \oplus \frac{v_O}{\sqrt{2}}. \end{aligned}$$

In particular, neither Hadamard acts on the catalyst. Every query vector of the phase transducer is scaled by $1/\sqrt{2}$, halving its cost. Replacing each controlled R_O by its two oracle calls doubles the cost, since the intervening known reflection preserves the queried norm. The factors cancel, giving

$$L_{\mathcal{T}}(O) = \frac{1}{2} \cdot 2 \cdot \frac{1}{|1-2p_O|} = \frac{1}{2|p_O-1/2|} \leq \frac{1}{2\delta}. \quad (3.10)$$

The chosen catalyst $w_O = v_O/\sqrt{2}$ satisfies

$$\|w_O\|^2 = \frac{\min\{p_O, 1 - p_O\}}{2|1 - 2p_O|} \leq \frac{1 - 2\delta}{8\delta}. \quad (3.11)$$

This proves the unrestricted upper bound. The circuit does not use δ , and (3.11) gives transduction complexity $O(1/\delta)$ on the promised family. One application contains four controlled query gates; the charged resource is their total queried squared norm in (3.10).

3.2 A matching four-oracle lower bound

Set

$$\ell = \frac{1}{2} - \delta, \quad h = \frac{1}{2} + \delta, \quad \mu = 2\sqrt{\ell h} = \sqrt{1 - 4\delta^2}. \quad (3.12)$$

Choose $\theta_L, \theta_H \in (0, \pi/2)$ by

$$(\cos \theta_L, \sin \theta_L) = (\sqrt{h}, \sqrt{\ell}), \quad (\cos \theta_H, \sin \theta_H) = (\sqrt{\ell}, \sqrt{h}).$$

Their difference $\Delta = \theta_H - \theta_L$ satisfies

$$\cos \Delta = \mu, \quad \sin \Delta = 2\delta. \quad (3.13)$$

A two-instance adversary using only the boundary rotations has filter norm $2 \sin(\Delta/2)$, yielding only $1/[2 \sin(\Delta/2)] < 1/(2\delta)$. Pairing each rotation with its negative gives the sharper joint filter below.

On $M_0 = \text{span}\{|0_A 0_W\rangle, |1_A 0_W\rangle\}$, define four oracles by

$$O_{c,\varepsilon}|_{M_0} = \varepsilon V(\theta_c), \quad V(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad c \in \{L, H\}, \quad \varepsilon \in \{+1, -1\}. \quad (3.14)$$

All four act as the identity on $M_0^\perp$. They are legal inputs for every W : their answer probabilities are ℓ or h , and their signs can be absorbed into the garbage states. Let $Q_{c,\varepsilon} = O_{c,\varepsilon} \oplus O_{c,\varepsilon}^*$. When $\dim W = 1$, the signs in (3.14) are global oracle signs, which are legitimate in the controlled-unitary convention fixed in (2.3).

Write $x = (c_x, \varepsilon_x)$. The public Gram difference is $F_{xy} = \mathbf{1}_{\{c_x \neq c_y\}}$. Let v_x be the total query vectors of any exact purifier on these four instances, and put $C = \max_x \|v_x\|^2$. Order the instances as $(L, +), (L, -), (H, +), (H, -)$, and set

$$\alpha = \frac{1 + \mu}{2}, \quad \beta = \frac{1 - \mu}{2}, \quad X_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K = \alpha I_2 + \beta X_2, \quad \Gamma = \begin{pmatrix} 0 & K \\ K & 0 \end{pmatrix}. \quad (3.15)$$

Each row of K sums to one, so

$$\sum_{x,y} \Gamma_{xy} F_{xy} = 4. \quad (3.16)$$

We compute the norm of the Hermitian filtered operator

$$D = [\Gamma_{xy}(I - Q_x^* Q_y)]_{x,y}. \quad (3.17)$$

The active rotations have a common complex eigenbasis. In either forward eigenchannel, put $z = e^{\pm i\Delta}$. The low-to-high block of D on the two sign indices is

$$C_z = \alpha(1 - z)I_2 + \beta(1 + z)X_2. \quad (3.18)$$

It is diagonalized by the Hadamard matrix, with eigenvalues $1 - \mu z$ and $\mu - z$. Since $\operatorname{Re} z = \mu$,

$$|1 - \mu z|^2 = |\mu - z|^2 = 1 - \mu^2 = 4\delta^2,$$

and therefore $\|C_z\| = 2\delta$. The high-to-low block is C_z^* . The square of the corresponding Hermitian block $\begin{pmatrix} 0 & C_z \\ C_z^* & 0 \end{pmatrix}$ is $\operatorname{diag}(C_z C_z^*, C_z^* C_z)$, so the Hermitian block has norm 2δ . The inverse direction replaces z by $\bar{z}$, and the common identity action on $M_0^\perp$ contributes zero. Ancillary and query-time controls only amplify these operators by an identity. Hence

$$\|D\| = 2\delta. \quad (3.19)$$

Multiplying (2.10) by Γ_{xy} , summing over the four instances, and applying (3.19) gives

$$4 = \left\langle \bigoplus_x v_x, D \bigoplus_x v_x \right\rangle \leq 2\delta \sum_x \|v_x\|^2 \leq 8\delta C.$$

Thus $C \geq 1/(2\delta)$. The four-instance family embeds into every nonzero workspace, so this lower bound and (3.10) prove (1.1).

4 Queries on the prescribed rays

4.1 An exact signed-Gram characterization

Index the full promised family by x , and write $\phi_x = O_x s$, $p_x = \|\Pi_1 \phi_x\|^2$, and $r_x = r_{O_x}$. Its public Gram difference is

$$F_{xy} = 1 - \delta_{r_x, r_y} = \mathbf{1}_{\{r_x \neq r_y\}}.$$

The restriction to rays turns the operator-valued adversary identity into a scalar kernel identity. The inverse direction contributes the negative of the forward filter; retaining this sign is essential.

Proposition 4.1 (Admissible adversary). *The value $C_{\text{adm}, W}(\delta)$ is the infimum of*

$$\sup_x (\|f_x\|^2 + \|g_x\|^2) \quad (4.1)$$

over fixed Hilbert spaces $\mathcal{K}_{\text{fwd}}, \mathcal{K}_{\text{inv}}$ and families $f_x \in \mathcal{K}_{\text{fwd}}$, $g_x \in \mathcal{K}_{\text{inv}}$ satisfying

$$F_{xy} = (1 - \langle \phi_x, \phi_y \rangle)(\langle f_x, f_y \rangle - \langle g_x, g_y \rangle) \quad (x, y). \quad (4.2)$$

Each feasible family is realized by one fixed admissible transducer with these pointwise query costs.

Proof. For an admissible query vector as in (2.6), the forward contribution to (2.10) is

$$\langle f_x, f_y \rangle (1 - \langle O_x s, O_y s \rangle) = \langle f_x, f_y \rangle (1 - \langle \phi_x, \phi_y \rangle).$$

In the inverse direction,

$$\langle \phi_x, (I - O_x O_y^*) \phi_y \rangle = \langle \phi_x, \phi_y \rangle - \langle O_x^* \phi_x, O_y^* \phi_y \rangle = \langle \phi_x, \phi_y \rangle - 1.$$

Adding the two contributions proves (4.2), and the cost is $\|f_x\|^2 + \|g_x\|^2$.

Conversely, set

$$v_x = (f_x \otimes s)_{\text{fwd}} \oplus (g_x \otimes \phi_x)_{\text{inv}}.$$

Equation (4.2) is precisely the adversary identity for $\xi_0 \mapsto \xi_{r_x}$. The construction in (2.11) already realizes this conversion. One can also prescribe the full public bit flip without any additional cost. For $b, c \in \{0, 1\}$,

$$\langle \xi_b, \xi_c \rangle - \langle \xi_{b \oplus r_x}, \xi_{c \oplus r_y} \rangle = (-1)^{b+c} F_{xy}, \quad (4.3)$$

where $\oplus$ in bit indices denotes addition modulo two. Thus the two families indexed by (b, x) ,

$$a_{b,x} = \xi_b \oplus (-1)^b Q_x v_x, \quad b_{b,x} = \xi_{b \oplus r_x} \oplus (-1)^b v_x,$$

have equal Gram matrices. Applying Lemma 2.1 to these families gives a fixed work unitary. Preceding it by the query realizes $\xi_b \mapsto \xi_{b \oplus r_x}$ with catalyst $(-1)^b v_x \oplus 0$. For the required input $b = 0$, the catalyst is $v_x \oplus 0$ and the cost is $\|v_x\|^2$, as claimed. The added summand is non-query space, so query support is preserved. $\square$

In particular, the admissible optimization depends on an oracle only through its generated state. Its completion disappears from the signed-Gram constraint, even though the fixed transducer must work for every completion.

4.2 Solving the kernel constraint

For opposite labels, (4.2) asks for the reciprocal kernel $1/(1 - \langle \phi_x, \phi_y \rangle)$. Summing tensor powers of the unit vectors ϕ_x would give this kernel formally, but the witness vectors would have infinite norm. The needed modification is to contract the vectors without changing these cross-class inner products. We use the reciprocal rescaling from [1, proof of Theorem 35] and [3, proof of Theorem 13.1], keeping its exact norm bound. Retain ℓ, h, μ from (3.12), and set

$$\kappa = \left(\frac{\ell}{h} \right)^{1/4}.$$

For each instance, define

$$c_x = \begin{cases} \kappa \Pi_0 \phi_x + \kappa^{-1} \Pi_1 \phi_x, & r_x = 0, \\ \kappa^{-1} \Pi_0 \phi_x + \kappa \Pi_1 \phi_x, & r_x = 1. \end{cases} \quad (4.4)$$

For opposite labels, the rescalings cancel on each answer block, so

$$\langle c_x, c_y \rangle = \langle \phi_x, \phi_y \rangle \quad (r_x \neq r_y). \quad (4.5)$$

At the same time,

$$\|c_x\|^2 \leq \sqrt{\frac{\ell}{h}} h + \sqrt{\frac{h}{\ell}} \ell = 2\sqrt{\ell h} = \mu < 1. \quad (4.6)$$

Indeed, for a low instance, $\|c_x\|^2 = \kappa^2(1 - p_x) + \kappa^{-2}p_x$ is increasing in p_x , so its maximum on the promised interval is attained at $p_x = \ell$. For a high instance the coefficients are interchanged; the resulting expression decreases in p_x and is maximized at $p_x = h$. Both maxima equal μ .

Thus the rescaling preserves precisely the inner products that occur in the cross-class constraints and makes every vector strictly contractive. Its tensor powers can therefore be summed in the full tensor Fock space

$$\mathcal{F}(M) = \bigoplus_{k=0}^{\infty} M^{\otimes k}, \quad M^{\otimes 0} = \mathbb{C}.$$

Define

$$\eta_x = \bigoplus_{k=0}^{\infty} c_x^{\otimes k}. \quad (4.7)$$

The norm bound (4.6) makes this a well-defined vector with

$$\|\eta_x\|^2 = \frac{1}{1 - \|c_x\|^2} \leq \frac{1}{1 - \mu}. \quad (4.8)$$

Moreover, $|\langle c_x, c_y \rangle| \leq \mu < 1$, so its inner products are given by an absolutely convergent geometric series:

$$\langle \eta_x, \eta_y \rangle = \sum_{k=0}^{\infty} \langle c_x, c_y \rangle^k = \frac{1}{1 - \langle \phi_x, \phi_y \rangle} \quad (r_x \neq r_y). \quad (4.9)$$

This step is valid for complex inner products; no choice of real garbage states is involved.

Let $\sigma_x = (-1)^{r_x}$, and choose

$$f_x = \frac{\eta_x}{\sqrt{2}}, \quad g_x = \frac{\sigma_x \eta_x}{\sqrt{2}}. \quad (4.10)$$

For equal labels, the two Gram terms in (4.2) cancel. For opposite labels, their difference is $\langle \eta_x, \eta_y \rangle$, so (4.9) verifies the constraint. Consequently, Proposition 4.1 gives one fixed admissible transducer with

$$L_{\mathcal{T}}(O_x) = \|f_x\|^2 + \|g_x\|^2 = \frac{1}{1 - \|c_x\|^2} \leq \frac{1}{1 - \mu}. \quad (4.11)$$

The Fock space is separable because M is finite-dimensional, and the stabilization adds only a separable non-query summand. The chosen catalyst has the same squared norm as the query cost. Its dependence on the class label in (4.4) and (4.10) is permitted: the work unitary is constructed once from all the Gram relations, and no gate receives the unknown label.

For the lower bound, choose any unit vector $w \in W$ and consider

$$\phi_- = \sqrt{h} |0\rangle w + \sqrt{\ell} |1\rangle w, \quad \phi_+ = \sqrt{\ell} |0\rangle w + \sqrt{h} |1\rangle w. \quad (4.12)$$

Each can be the image of s under a unitary: extend s and the chosen image separately to orthonormal bases and map one basis to the other. These are therefore legal low and high instances, and $\langle \phi_-, \phi_+ \rangle = \mu$. Any feasible witnesses must satisfy

$$\frac{1}{1 - \mu} = |\langle f_-, f_+ \rangle - \langle g_-, g_+ \rangle| \leq \sqrt{(\|f_-\|^2 + \|g_-\|^2)(\|f_+\|^2 + \|g_+\|^2)},$$

where the inequality is Cauchy-Schwarz applied to (f_-, g_-) and $(f_+, -g_+)$. At least one cost is at least $1/(1 - \mu)$. Together with (4.11), this proves (1.2) and completes the proof of Theorem 1.1.

The opposite signs in (4.10) cancel every same-label constraint without restricting the within-class overlaps of the Fock vectors. This cancellation is essential. With only one query direction, distinct same-label inputs would require orthogonal witnesses, while each must retain a nonzero overlap with a fixed opposite-label witness.

Proposition 4.2. *For the full promised family, the uniform exact admissible cost is infinite if either all forward queries or all inverse queries are forbidden. This already holds for $\dim W = 1$.*

Proof. Consider first the forward-only case, so $g_x = 0$ in (4.2). For any positive integer N , choose distinct $p_1, \dots, p_N \in (0, \ell)$ and let

$$\phi_i = \sqrt{1 - p_i} |0\rangle + \sqrt{p_i} |1\rangle.$$

Fix a real high-probability state ϕ_+ of the same form. For $i \neq j$, the low states are distinct unit vectors, hence $1 - \langle \phi_i, \phi_j \rangle \neq 0$. The same-label constraints force $\langle f_i, f_j \rangle = 0$. The opposite-label constraints give

$$\langle f_i, f_+ \rangle = \frac{1}{1 - \langle \phi_i, \phi_+ \rangle}, \quad |\langle f_i, f_+ \rangle| \geq \frac{1}{2},$$

because $|1 - \langle \phi_i, \phi_+ \rangle| \leq 2$. In particular, every f_i is nonzero. If all squared norms were bounded by a finite $C > 0$, Bessel's inequality for the orthonormal vectors $f_i/\|f_i\|$ would give

$$C \geq \|f_+\|^2 \geq \sum_{i=1}^N \frac{|\langle f_i, f_+ \rangle|^2}{\|f_i\|^2} \geq \frac{N}{4C}.$$

Thus $C \geq \sqrt{N}/2$, which is impossible uniformly in N . In the inverse-only case, the same-label constraints make the g_i mutually orthogonal, while $\langle g_i, g_+ \rangle = -1/(1 - \langle \phi_i, \phi_+ \rangle)$. The same absolute-value and Bessel estimates apply. The two-dimensional family embeds into $A \otimes W$ by tensoring with a fixed unit vector of W , so the obstruction holds for every nonzero workspace. $\square$

The lower bounds use only finite subfamilies at a time and require no separability assumption. Allowing nonseparable private space therefore cannot lower either optimum, whereas the constructions attain both values on fixed separable spaces. Finite-dimensional approximate realizations of purification are treated in [3, 4].

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